



## Generating a sequence using a position-to-term rule

1 Match up the keywords to their definitions Write the correct letter in each box

|   |                       |
|---|-----------------------|
| a | term                  |
| b | term-to-term rule     |
| c | position-to-term rule |
| d | sequence              |

|          |                                                              |
|----------|--------------------------------------------------------------|
| <i>d</i> | a succession of values or patterns usually following a rule. |
| <i>c</i> | describes how to calculate the term from the term number     |
| <i>a</i> | each value or pattern in a sequence                          |
| <i>b</i> | describes how to get from one term to the next               |

2 Using the table shown, match the notation to the term. Write the correct letter in each box

|             |   |   |   |    |
|-------------|---|---|---|----|
| Term number | 1 | 2 | 3 | 4  |
| Term        | 3 | 1 | 0 | -2 |

|   |       |
|---|-------|
| a | $T_1$ |
| b | $T_2$ |
| c | $T_3$ |
| d | $T_4$ |

|          |    |
|----------|----|
| <i>c</i> | 0  |
| <i>d</i> | -2 |
| <i>a</i> | 3  |
| <i>b</i> | 1  |



- 3 The position-to-term rule for a sequence is 'multiply the term number by 5 then subtract 10'. What is the 20th term in this sequence? Fill in the blank

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- 4 Which of these is a position-to-term rule for the number of dots in each pattern of this sequence? Tick **1** correct answer



- ☐ add 6 to the previous term
- ☒ multiply the term number by 6 then add 4
- ☐ multiply the term number by 6 then add 10
- ☐ Multiply the term number by 10
- ☐ Multiply the term number by 10 then subtract 4

- 5 Which of these is a position-to-term rule for the sequence 12, 22, 32, 42, 52, .... Tick **1** correct answer

- ☐ the term number add 10
- ☐ multiply the term number by 10
- ☒ multiply the term number by 10 then add 2
- ☐ multiply the term number by 12
- ☐ multiply the term number by 12 and subtract 2



- 6 Work out the 100th term in the constant additive sequence which starts: 13, 21, 29, 37, ...  
The first few lines of working are shown. Fill in the blank

$$\begin{array}{ccccccc} 13, & 21, & 29, & 37, & \dots \\ \text{---} & \text{---} & \text{---} & & \\ & +8 & +8 & +8 & \end{array}$$

$$T_1 = 8 + 5$$

$$T_2 = 8 + 8 + 5$$

$$T_3 = 8 + 8 + 8 + 5$$

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